# A standardized workflow for computer-assisted language comparison

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31.08.2026

Language Science Meets Linguistic Diversity



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# Introduction

# Goals

## Content

- Present you a workflow that is easy to use for your own study

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  1. Data digitization

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  1. Data digitization
  2. The role of CLDF

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  3. Annotating comparative wordlists

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- Present you a workflow that is easy to use for your own study
  1. Data digitization
  2. The role of CLDF
  3. Annotating comparative wordlists
  4. Further steps

# Cross-Linguistic Data Formats (CLDF)

# CLDF




## Why?

To allow exchange of cross-linguistic data and decouple development of tools and methods from that of databases, standardized data formats are necessary.

Once established, these dataformats could become a foundation not only for tools but also for instruction material in the spirit of Data Carpentry for historical linguistics and linguistic typology.

## What?

The main types of cross-linguistic data we are concerned with here are any tabular data which is typically analysed using quantitative (automated) methods or made accessible using software tools like the `clld` framework, such as

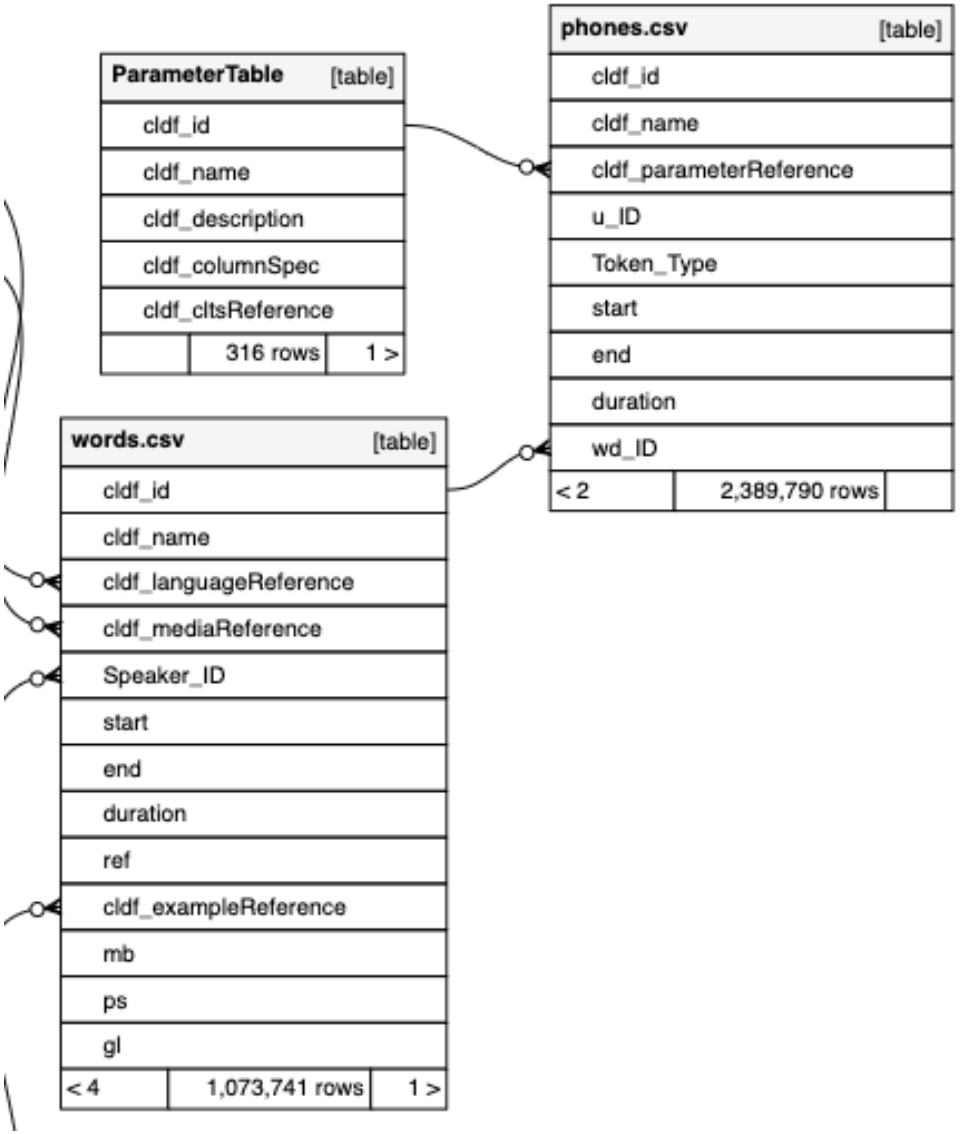
- wordlists (or more complex lexical data including e.g. cognate judgements),
- structure datasets (e.g. WALS features),
- simple dictionaries.

# CLDF

## Features

- One value per cell
- Multiple tables
- Standardization ensures comparability
- Validation through provided software

# CLDF

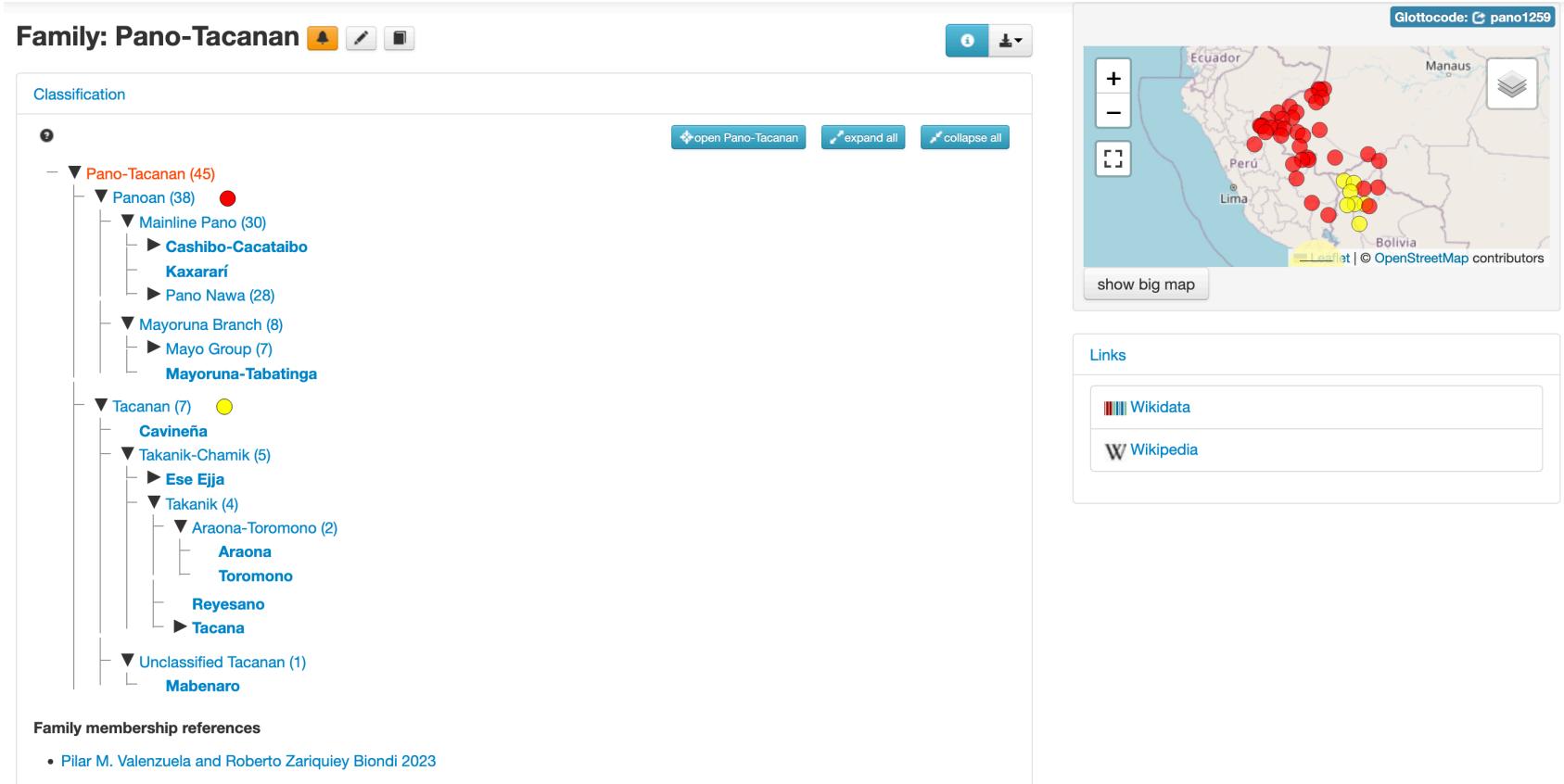

ParameterTable [table]
cldf_id
cldf_name
cldf_description
cldf_columnSpec
cldf_cltsReference
316 rows
1 >
phones.csv [table]
cldf_id
cldf_name
cldf_parameterReference
u_ID
Token_Type
start
end
duration
wd_ID
< 2
2,389,790 rows
words.csv [table]
cldf_id
cldf_name
cldf_languageReference
cldf_mediaReference
Speaker_ID
start
end
duration
ref
cldf_exampleReference
mb
ps
gl
< 4
1,073,741 rows
1 >

# CLDF

## Reference catalogues

- Glottolog
- CLTS
- Concepticon

# Glottolog for languages (Hammarström et al. 2026)



# CLTS for phonetic features (List et al. 2024)

## Sound *voiceless post-alveolar sibilant affricate consonant*

Showing 1 to 36 of 36 entries

← Previous | 1 | Next →

| Grapheme | Dataset | Type |
|---|---|---|
| Search | Search | --any-- |
| C | asjpcode | transcription system |
| tʃ | bipa | transcription system |
| t͡ʃ | bipa | transcription system |
| tʲ | bipa | transcription system |
| ʧ | bipa | transcription system |
| č | gld | transcription system |
| tʲ | napa | transcription system |
| tʃ | allenbai | transcription data |
| t͡ʃ | apics | transcription data |
| tʃ | bdpa | transcription data |
| tʃ | bdproto | transcription data |
| tʃ | bdproto | transcription data |
| t͡ʃ | bdproto | transcription data |
| ʧ | bdproto | transcription data |
| tʃ | beidasinitic | transcription data |
| č | chomsky | transcription data |
| t͡ʃ | easterday | transcription data |
| tʃ | eurasian | transcription data |

**Grapheme**
tʃ
**Unicode**
LATIN SMALL LETTER T / LATIN SMALL LETTER ESH
**Category**
consonant
**Features**
airstream: sibilant
place: post-alveolar
phonation: voiceless
manner: affricate

# Concepticon for conceptlists (List et al. 2025)

## Concept lists

Showing 1 to 13 of 13 entries (filtered from 466 total entries)

← Previous | 1 | Next →

| Note | Name | Compiler | Alias | Items | Tags | Uniqueness | Year | Gloss languages | Target languages | Sources |
|---|---|---|---|---|---|---|---|---|---|---|
| | swades | | Search | Search | --any-- | Search | Search | --any-- | Search | |
| more | Swadesh 1952 200 | Swadesh, Morris | Swadesh-200 | 200 | basic | 0.01 | 1952 | english | Global | Swadesh 1952 |
| more | Swadesh 1964 100 | Swadesh, Morris | Swadesh-100-Russian | 100 | basic | 0.00 | 1964 | english russian | Global | Swadesh 1964 |
| more | Swadesh 1955 100 | Swadesh, Morris | Swadesh-100 | 100 | basic | 0.00 | 1955 | english | Global | Swadesh 1955 |
| more | Swadesh 1960 215 | Swadesh, Morris | | 215 | basic | 0.01 | 1960 | english russian | Global | Swadesh 1964 |
| more | Swadesh 1954 97 | Swadesh, Morris | | 97 | basic | 0.00 | 1954 | english | American Indian languages | Swadesh 1954 |
| more | Swadesh 1971 100 | Swadesh, Morris | | 100 | basic | 0.00 | 1971 | english | Global | Swadesh 1971 |
| more | Swadesh 1960 100a | Swadesh, Morris | | 100 | basic | 0.00 | 1960 | english | Global | Hymes 1960 |
| more | Swadesh 1960 200 | Swadesh, Morris | | 200 | basic | 0.01 | 1960 | english | Global | Hymes 1960 |
| more | Swadesh 1950 215 | Swadesh, Morris | | 215 | basic | 0.01 | 1950 | english | Global | Swadesh 1950 |
| more | Swadesh 1955 215 | Swadesh, Morris | | 215 | ranked | 0.01 | 1955 | english | Global | Swadesh 1955 |
| more | Swadesh 1960 100b | Swadesh, Morris | | 100 | unstable | 0.01 | 1960 | english | Global | Hymes 1960 |
| more | Swadesh 1950 165 | Swadesh, Morris | | 165 | areal | 0.01 | 1950 | english | Salish | Swadesh 1950 |

# Concepticon for concepts

Showing 1 to 100 of 321 entries

← Previous 1 2 3 4 Next →

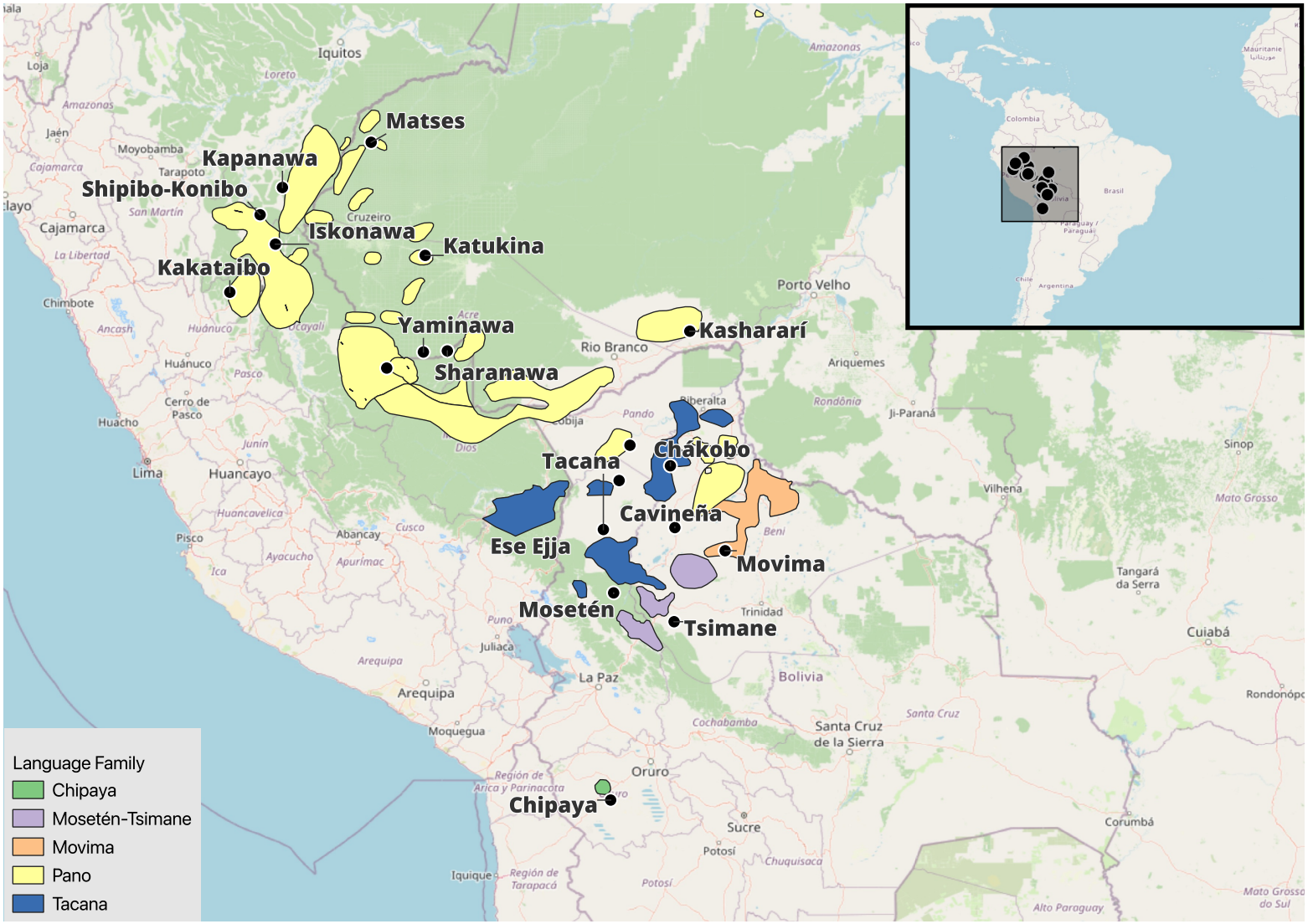
| Id | Concept in source | Conceptlist |
|---|---|---|
| Search | Search | Search |
| Abraham-2018-307-87 | fire [english] | Abraham 2018 307 |
| Allen-2007-500-24 | 火 [chinese]; fire [english] | Allen 2007 500 |
| Alpher-1999-151-29 | fire [english] | Alpher 1999 151 |
| Anonby-2018-1500-1112 | fire [english]; آتش [persian] | Anonby 2018 1500 |
| Araujo-1996-289-69 | Feuer [german]; fogo [portuguese] | Araujo 1996 289 |
| Ardila-2007-40-14 | Su [basque]; Fire [english]; Fajro [esperanto]; Iso [guahibo]; Ignis [latin]; Ogon [russian]; Fuego [spanish]; Ates [turkish] | Ardila 2007 40 |
| Auderset-2023-240-134 | FIRE [english] | Auderset 2023 240 |
| Backstrom-1992-210a-55 | fire [english] | Backstrom 1992 210a |
| Backstrom-1992-210b-55 | fire [english] | Backstrom 1992 210b |
| Backstrom-1992-210c-55 | fire [english] | Backstrom 1992 210c |
| Backstrom-1992-210d-55 | fire [english] | Backstrom 1992 210d |
| Backstrom-1992-210e-55 | fire [english] | Backstrom 1992 210e |
| Backstrom-1992-210f-55 | fire [english] | Backstrom 1992 210f |
| Bastin-1999-92-27 | feu [french] | Bastin 1999 92 |
| Bender-1971-99-27 | fire [english] | Bender 1971 99 |
| Benedict-1976-100-82 | fire [english] | Benedict 1976 100 |

# Adding data from other datasets: Grambank

```
> attach 'blumpanotacana.sqlite' AS db1;
> attach 'grambank.sqlite' AS db2;

> SELECT *
> FROM db1.LanguageTable AS a
> INNER JOIN db2.ValueTable AS b
> ON a.cldf_glottocode = b.cldf_languageReference;
```

# Adding data from other datasets: Language polygons



# CLDF datasets: DoReCo

Corpus files : download languages metadata download citations in bibtex

Showing 1 to 53 of 53 entries

← Previous 1 Next →

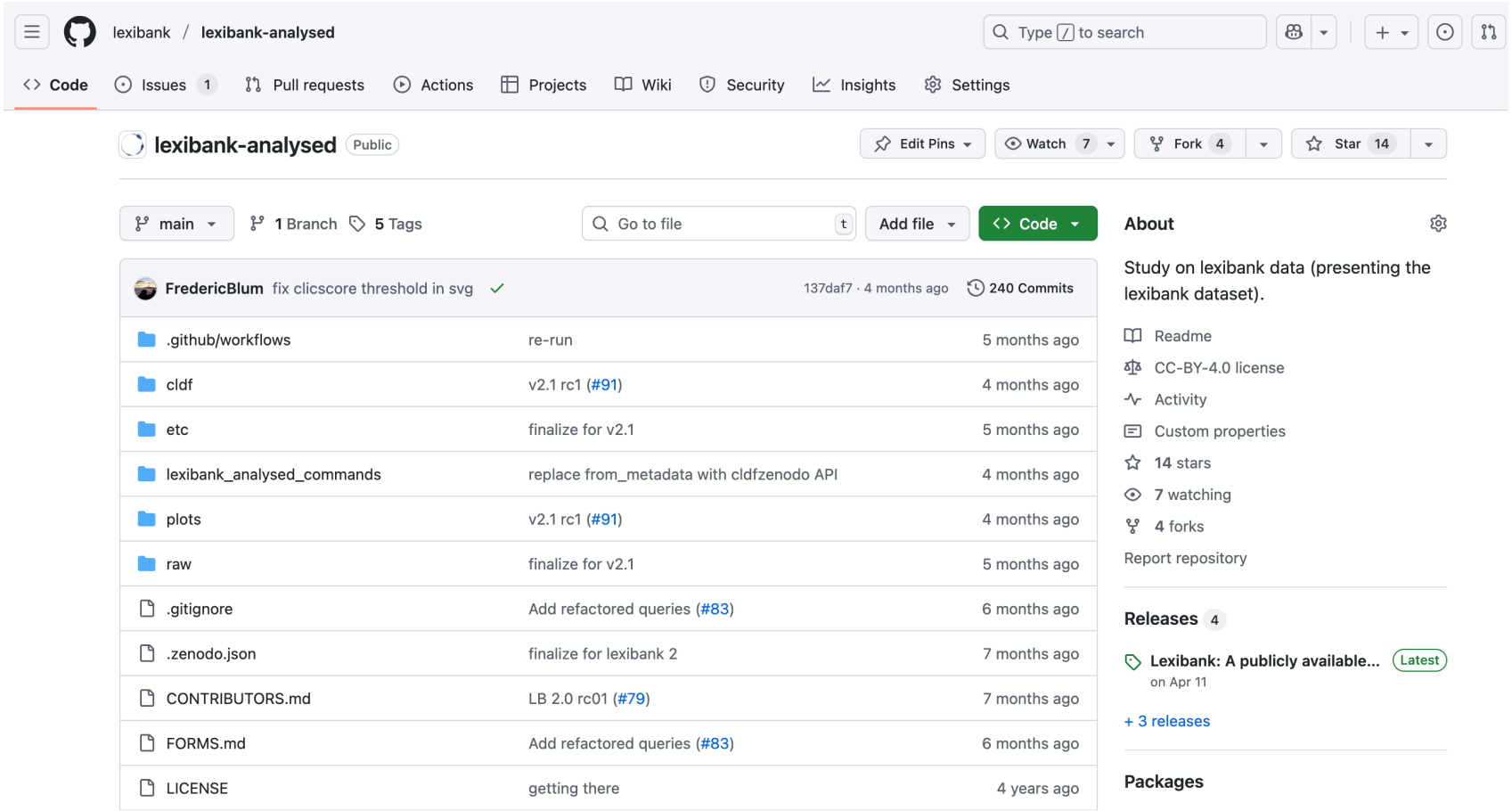
| Language | Glottocode | Family | Area | Creator(s) | License(s) | Word tokens | Speakers | Texts | Gloss |
|---|---|---|---|---|---|---|---|---|---|
| Anal | anal1239 | Sino-Tibetan | Eurasia | Ozerov, Pavel | CC BY, CC BY-NC | 13190 | 12 | 23 | none |
| Arapaho | arap1274 | Algic | North America | Cowell, Andrew | CC BY | 4612 | 4 | 11 | all |
| Asimjeeg Datooga | tsim1256 | Nilotic | Africa | Griscom, Richard | CC BY | 8782 | 12 | 40 | none |
| Baïnounk Gubëeher | bain1259 | Atlantic-Congo | Africa | Cobbinah, Alexander Yao | CC BY | 11423 | 9 | 17 | some |
| Beja | beja1238 | Afro-Asiatic | Africa | Vanhove, Martine | CC BY-NC | 15427 | 5 | 58 | all |
| Bora | bora1263 | Boran | South America | Seifart, Frank | CC BY | 8364 | 6 | 9 | all |
| Cabécar | cabe1245 | Chibchan | North America | Quesada, Juan Diego and Skopeteas, Stavros and Pasamonik, Carolina and Brokmann, Carolin and Fischer, Florian | CC BY-NC, CC BY-NC-ND | 10613 | 10 | 39 | all |
| Cashinahua | cash1254 | Pano-Tacanan | South America | Reiter, Sabine | CC BY | 9655 | 3 | 7 | some |
| Daakie | port1286 | Austronesian | Papunesia | Krifka, Manfred | CC BY | 11880 | 10 | 16 | some |
| Dalabon | ngal1292 | Gunwinyguan | Australia | Ponsonnet, Maïa | CC BY | 3340 | 4 | 10 | some |
| Dolgan | dolg1241 | Turkic | Eurasia | Däbritz, Chris Lasse and Kudryakova, Nina and Stapert, Eugénie and Arkhipov, Alexandre | CC BY, CC BY-NC | 8775 | 6 | 9 | all |
| English (Southern England) | sout3282 | Indo-European | Eurasia | Schiborr, Nils Norman | CC BY | 8914 | 2 | 4 | all |
| Evenki | even1259 | Tungusic | Eurasia | Kazakevich, Olga and Klyachko, Elena | CC BY | 8310 | 23 | 36 | all |
| Fanbyak | orko1234 | Austronesian | Papunesia | Franjieh, Michael | CC BY | 10162 | 10 | 26 | some |

LexiBank
(Blum et al. 2025; List et al. 2022)

# A collection of datasets

| Id | Name | # languages | # concepts | # words |
|---|---|---|---|---|
| Search | Search | Search | Search | Search |
| aaleykusunda | CLDF dataset derived from Aaley and Bodt's "New Kusunda data: A list of 250 concepts" from 2020 | 1 | 227 | 227 |
| abrahammonpa | CLDF dataset derived from Abraham et al.'s "Sociolinguistic Research on Monpa" from 2018 | 12 | 304 | 3,208 |
| allenbai | CLDF dataset derived from Allen's "Bai Dialect Survey" from 2007 | 9 | 499 | 4,546 |
| backstromnorthernpakistan | CLDF dataset derived from Backstrom and Radloff's "Sociolinguistic Survey of Northern Pakistan" from 1992 | 5 | 219 | 993 |
| bantubvd | CLDF dataset derived from Greenhill and Gray's "Bantu Basic Vocabulary Database" from 2015 | 10 | 415 | 4,187 |
| beidasinitic | CLDF dataset derived from Beijing University's "Chinese Dialect Vocabularies" from 1964 | 18 | 736 | 14,745 |
| birchallchapacuran | CLDF dataset derived from Birchall et al.'s "A Combined Comparative and Phylogenetic Analysis of the Chapacuran Language Family" from 2016 | 3 | 124 | 355 |
| blustaustronesian | CLDF dataset derived from Blust's Austronesian data coded for the Austronesian Basic Vocabulary Database from 2008 | 401 | 210 | 83,873 |
| bodtkhobwa | CLDF dataset derived from Bodt's "Lexical Cognates in Western Kho-Bwa" from 2019 | 8 | 542 | 4,264 |
| bowernpny | CLDF dataset derived from Bowern and Atkinson's "Internal Structure of Pama-Nyungan" from 2012 | 147 | 338 | 36,513 |
| bremerberta | CLDF dataset derived from Bremer's "Sociolinguistic Survey of Six Berta Speech Varieties in Ethiopia" from 2016 | 3 | 183 | 558 |

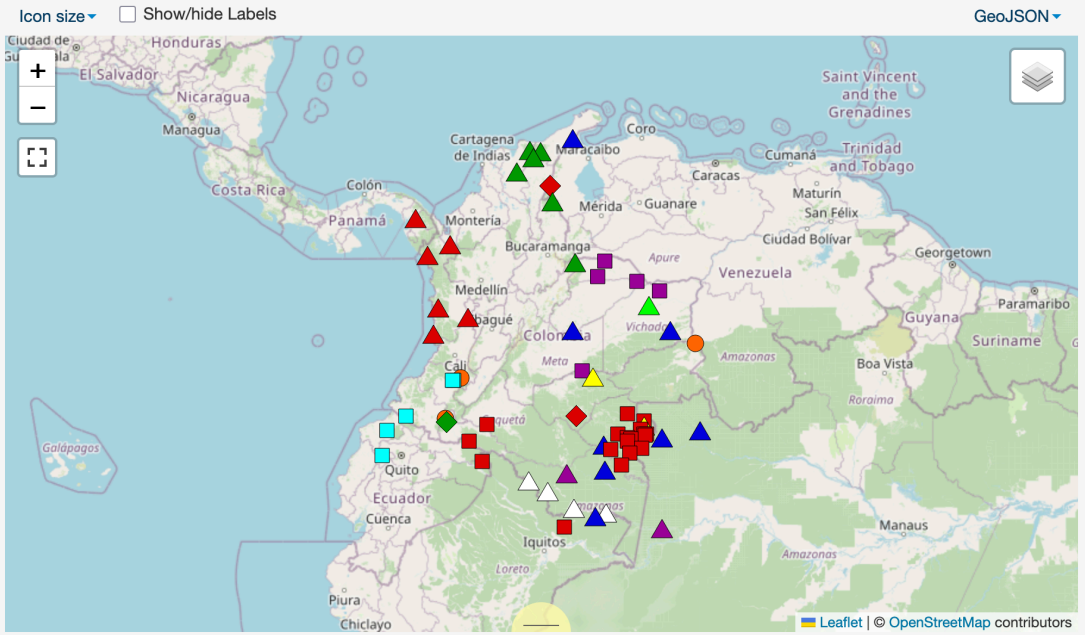
# Availability of data



# LexiBank datasets

## Dataset hubercolumbian

Varieties | Words




## CLDF Dataset

CLDF dataset derived from Huber and Reed's "Comparative Vocabulary" from 1992

**Zenodo:**
10.5281/zenodo.5121219
**GitHub:**
lexibank/hubercolumbian [v4.0.1]
**Source:**

> Huber, R. Q. and Reed, R. B. 1992. Vocabulario comparativo: palabras selectas de lenguas indígenas de Colombia [Comparative vocabulary. Selected words from the indigeneous languages of Columbia]. Santafé de Bogota: Asociatión Instituto Lingüístico de Verano.

**Conceptlists:**
Huber-1992-375

# LexiBank data

| Form | Concept | Segments | Variety | Family |
|---|---|---|---|---|
| Search | Search | Search | Search | --any-- |
| a | HAIR (HEAD) | a | Chimila | Chibchan |
| a | HE | a | Ika | Chibchan |
| a | SHE | a | Ika | Chibchan |
| a | SEED | a | Orejón | Tucanoan |
| a | HAIR (HEAD) | a | Tsafiqui Pila | Barbacoan |
| a | HAIR (BODY) | a | Tsafiqui Pila | Barbacoan |
| a | IF | a | Wanano | Tucanoan |
| a¹kaji² | FISH | a k a j i | Nukak | Kakua-Nukak |
| aa | COME | aː | Ocaina | Huitotoan |
| aá | WALK | a a | Tatuyo | Tucanoan |
| áá | WALK | a a | Carapana | Tucanoan |
| áá | THEY | a a | Ocaina | Huitotoan |
| ãã | SAY | ã ã | Ocaina | Huitotoan |
| ãã | DOWN | ã ã | Ocaina | Huitotoan |
| ãã́ | NOT | ã ã | Ocaina | Huitotoan |
| aaa | RIPE | aː a | Ocaina | Huitotoan |
| áaáamo | SHE | a a a a m o | Muinane | Boran |
| aabare | LONG | aː b a r e | Cha'palaachi | Barbacoan |
| áabata | OTHER | a a b a t a | Achagua | Arawakan |

# Data digitization (Oliveira 2014)

5. *ʔano ‘paca (*Cuniculus paca*)’ : Ksh ʔano : SK ʔano : Kp ʔano : Mar -- : Ch ʔano ‘jochi pintado’ (ZINGG, 1998) : Kax anawɨ : Yam ano, anõ ‘majás’ : Chan -- : Shar ano, anon ‘majás’ : Shan anu : Kat ano : Poy -- : A ʔanó ‘majás, roedor pequeño’ (HYDE, 1980), aˈno I : Kn anu : M ano ‘paca, clase de roedor’ : Yaw anu : Ko -- : Mt -- : My --.

# Data digitization (Blum & Barrientos 2023)

| DOCULECT | CONCEPT | ENTRY IN SOURCE | ALIGNMENT | PROTO SET |
|---|---|---|---|---|
| Proto-Panoan | paca | *?ano ‘paca (Cuniculus paca)’ | ? a n o - - - | §5 |
| Amawaka | paca | A ?anó ‘majás;; roedor pequeño’ (HYDE;;… | ? a n !ó o - - - | §5 |
| Amawaka | paca | A ?anó ‘majás;; roedor pequeño’ (HYDE;;… | ! ? a n o - - - | §5 |
| Chakobo | paca | Ch ?ano ‘jochi pintado’ (ZINGG;; 1998) | ? a n o - - - | §5 |
| Katukina | paca | Kat ano | - a n o - - - | §5 |
| Kaxarari | paca | Kax anawɨ | - a n a w ɨ - | §5 |
| Kaxinawa | paca | Kn anu | - a n u - - - | §5 |

## . . . in practice

What can I do with that data now?

# Colexifications in LexiBank

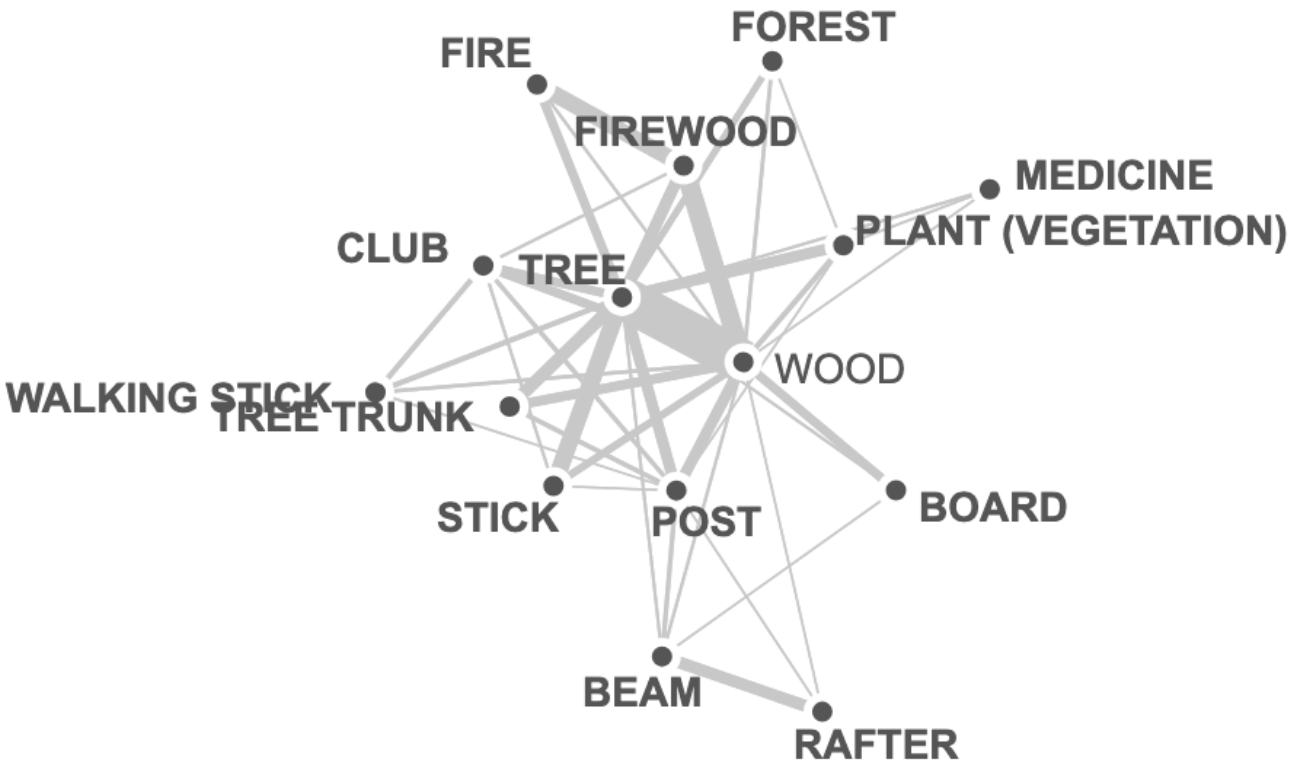



Figure 1: Colexification network inferred from LexiBank dataset.

# Conclusion

# Why should I do all this?

## Advantages

- Higher quality of data
- Accessible data that is readable for others
- Comparability of data ensures interoperability
- Large datasets for quantitive or comparative studies

# References I


Blum, Frederic & Carlos Barrientos. 2023. A new dataset with phonological reconstructions in CLDF. *Computer-Assisted Language Comparison in Practice* 6(1). 43–51. https://doi.org/10.15475/calcip.2023.1.6.

Blum, Frederic et al. 2025. Lexibank 2: pre-computed features for large-scale lexical data. *Open Research Europe* 5(126). 1–27. https://doi.org/10.12688/openreseurope.20216.2.

Hammarström, Harald et al. 2026. *Glottolog database (v5.3)*. Leipzig: Max Planck Institute for Evolutionary Anthropology. https://doi.org/10.5281/zenodo.18840935.

List, Johann-Mattis et al. 2022. Lexibank, a public repository of standardized wordlists with computed phonological and lexical features. *Scientific Data* 9(1). 1–16. https://doi.org/10.1038/s41597-022-01432-0.

List, Johann-Mattis et al. 2024. *CLTS. Cross-Linguistic Transcription Systems (v2.3.0)*. https://doi.org/10.5281/zenodo.10997741.

List, Johann-Mattis et al. 2025. *CLLD Concepticon (v3.4.0)*. Leipzig: Max Planck Institute for Evolutionary Anthropology. https://doi.org/10.5281/zenodo.14923561.

Oliveira, Sanderson Castro Soares de. 2014. *Contribuições para a reconstrução do Protopáno*. Brasília. https://repositorio.unb.br/handle/10482/17129.